ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Exercise: Filtering

Image Processing & Analysis for Life Scientists

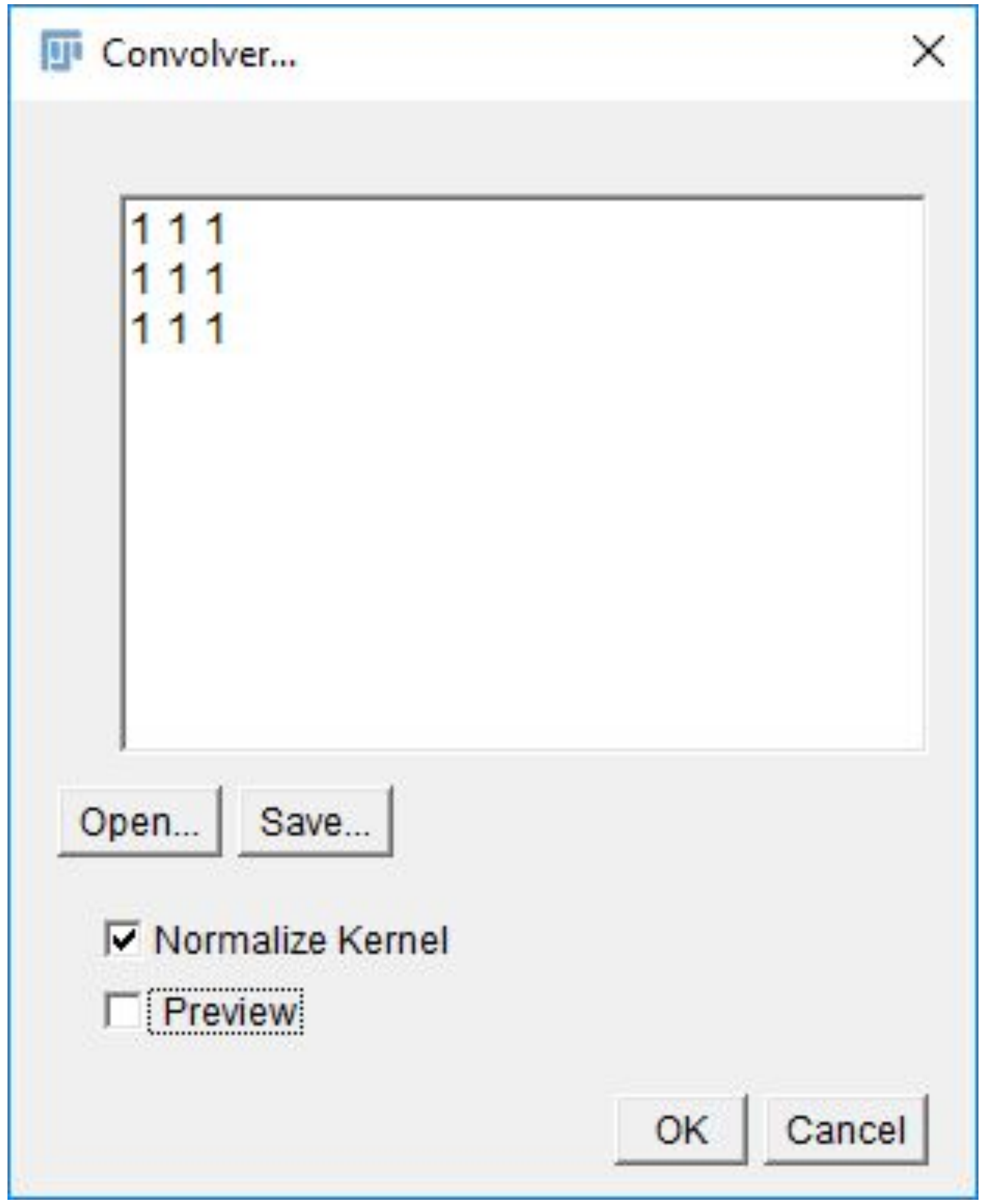
Olivier Burri, Romain Guiet & Arne Seitz

Spatial Filtering: Convolution

1. Open ***blobs.gif***
2. Zoom in till you see pixels
3. Launch
   Process→Filter→Convolve
   Match the kernel to the examples and use 'preview'

3x3

$$\begin{matrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{matrix}$$



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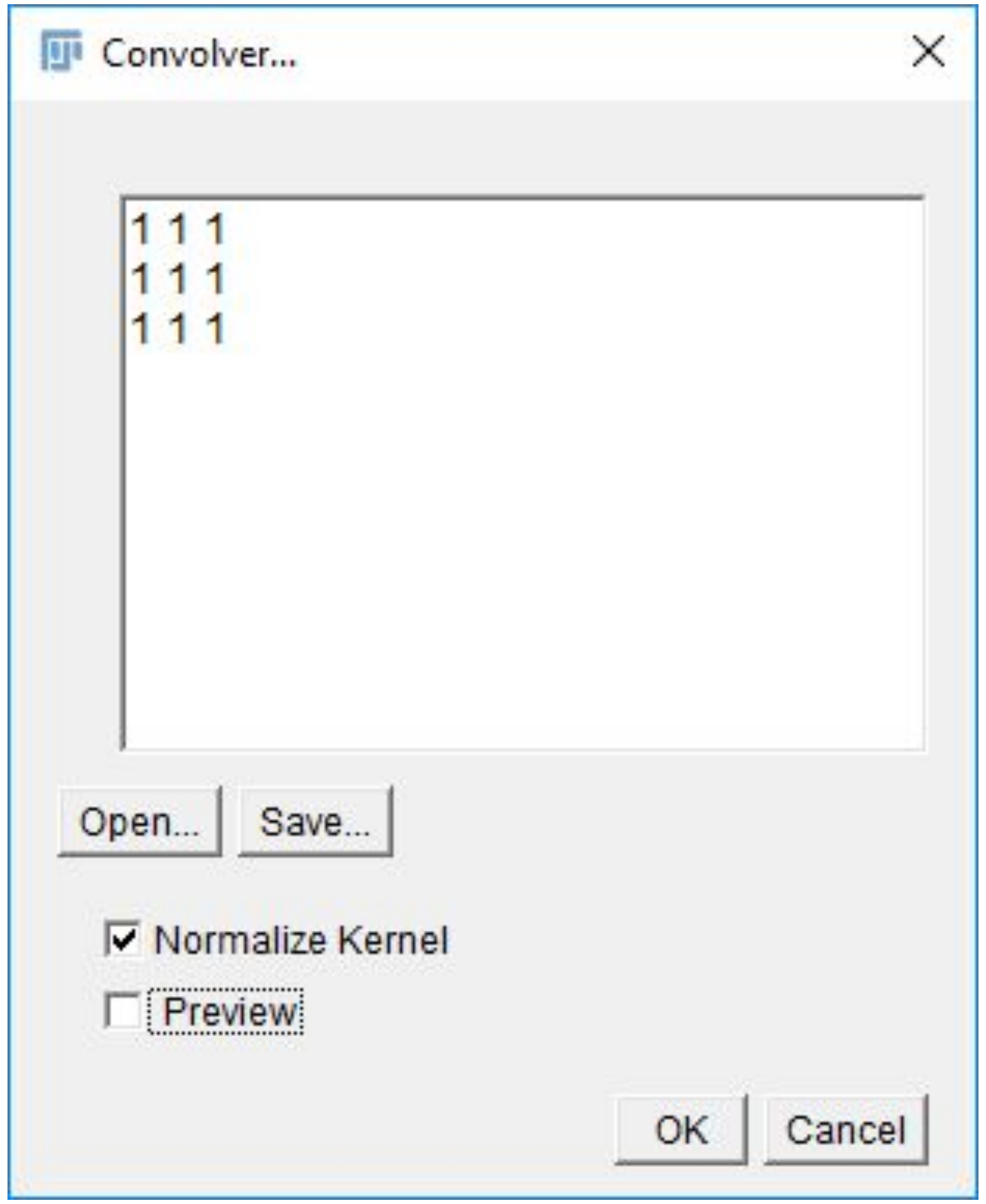
4. What happens if you uncheck "Normalize Kernel"?



3x3

1	1	1
1	1	1
1	1	1

5x5

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

9x9

1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1
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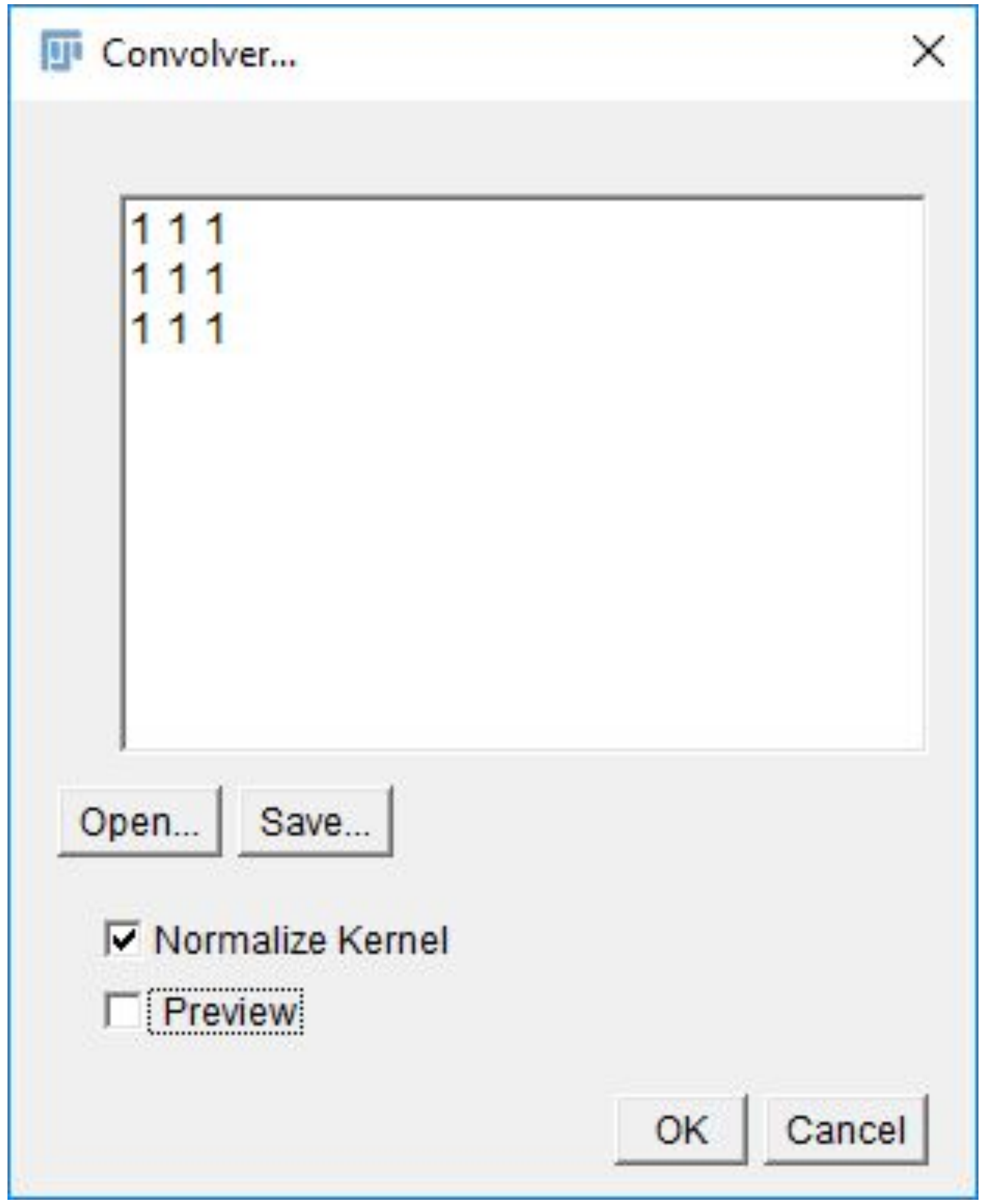
5. Try the **non-linear filter** 'Median'
   Process→Filter→Median



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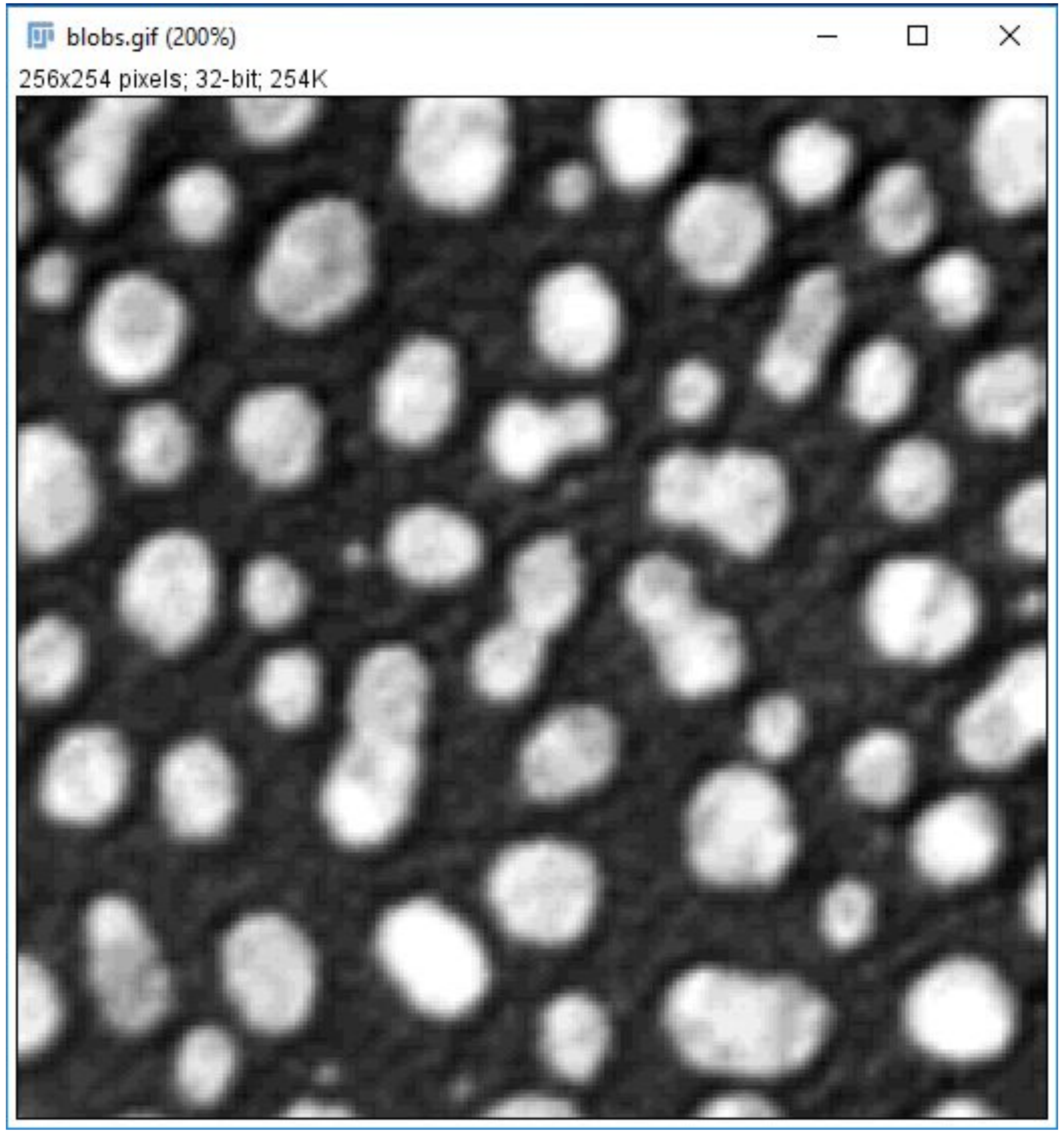
Convolution Example: Edge Detection

1. Open **Blobs** and transform it to 32-bit.
   Pay attention to the LUT.
2. Duplicate the image twice
3. On one duplicated image use the following filter:
   $$\begin{matrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{matrix}$$
4. On the other one, use:
   $$\begin{matrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{matrix}$$



Convolution Example: Edge Detection

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$$\boldsymbol{Result} = \sqrt{\mathrm{Image}_A^2 + \mathrm{Image}_B^2}$$

5. Calculate the square of the two images:
   Process→Math→Square *for each image*
6. Calculate the sum of the processed images.
   Process→Image Calculator…
7. *Merge the result with the original image:*
   Image→Color→Merge Channels…
8. Adjust the Brightness for optimal visualization.

Convolution Example: Edge Detection

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